

Optimum fuzzy sliding mode control of fuel sloshing in a spacecraft using PSO algorithm

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Abstract

This paper introduces an optimal fuzzy sliding mode (OFSM) controller for the control of fuel sloshing in a spacecraft during orbital maneuvers. In order to optimize controller's gains for minimizing control inputs, a Particle Swarm Optimization (PSO) algorithm is employed. A novel gain scheduling of the OFSM controller is developed for the coupled spacecraft-fuel system. Fuzzy gains are produced on-line based-on rules that utilize dynamic system variables and the sign of saturation function of sliding surfaces. The fuel slosh is modeled as a fixed spherical mass and two spherical pendulum masses moving in a spherical tank. Computer simulations illustrate the effectiveness of the proposed controller in maneuvering spacecraft while suppressing fuel sloshing. The performance of the OFSM controller is compared to that of the optimum State-Dependent Riccati Equation (SDRE) controller.

Keywords: Fuel slosh model, Nonlinear model, Attitude control, Optimum controller, Fuzzy controller, Gain scheduling

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1. Introduction

In control problems, in order to reduce settling time or control effort, it is essential to tune and optimize the controller's gains. In the last few decades, quite a few approaches have been developed to achieve these goals, for instance, fuzzy approaches [1], PSO [2] and bee colony [3] methods could be mentioned here. Among these approaches the fuzzy and the PSO are the best-known commonly employed methods. Optimization is very important in dynamic problems with high sensitivity and expensive control inputs as in spacecraft control systems, especially during high-thrust maneuvers [4, 5].

During space maneuvers attitude dynamics of spacecraft could be affected by a phenomenon called fuel sloshing which is caused by axial or rotational changes in spacecraft acceleration. Coupling between slosh dynamics and attitude dynamics can disturb the attitude of the spacecraft. Importance of this crucial problem make it an active research field in the last few decades [6–12]. For instance, in [13] equivalent single and multi mass-spring models of sloshing in a spacecraft was proposed. Whereas in [14, 15], single mass-spring and planar pendulum models were used to characterize the most prominent sloshing mode; in these two studies Lyapunov-based feedback controllers were employed to suppress sloshing and stabilize attitude of spacecraft simultaneously. In [16], a linearized form of a spacecraft dynamic model with fuel sloshing using a single pendulum model was derived, and an adaptive pole placement controller was designed for stabilization and tracking objectives. In [17], to characterize the most prominent two sloshing modes, a two-pendulum model was introduced and a Lyapunov-based controller was designed. A three dimensional modeling of sloshing using single mass-spring model was investigated in [18]. In addition to aforementioned equivalent models and control approaches employed for modeling and control of fuel sloshing in spacecraft, there are a number of other modeling and control approaches such as H_∞ controller [19], neural network creating an adaptive nonlinear dynamic inversion controller [20], linear quadratic regulator and linear quadratic Gaussian control methods [21], sliding

mode controller for the system with uncertainty [22], pulsating ball model with feedback control strategy based-on Lyapunov theory [23], an improved spherical pendulum model with Lyapunov-based output-feedback controller [24], and single pendulum model in three dimensional space with fuzzy parallel disturbed compensation and fuzzy linear quadratic regulator controllers [25].

This paper considers a full nonlinear coupled spacecraft-fuel system model, where the fuel slosh dynamics are described by two spherical pendulums as in [26]. In order to control the coupled spacecraft-fuel system, an efficient optimal fuzzy sliding mode (OFSM) controller is introduced. The proposed OFSM controller is defined by a novel gain scheduling scheme, where the fuzzy gains are generated on-line based on the dynamic system variables and the sign of saturation function of sliding surfaces. A PSO algorithm is employed to optimize fuzzy gains in order to achieve optimal controller performance with minimum control inputs. A simulation is conducted to demonstrate the effectiveness of the control approach. Results of simulation and comparison between the SDRE and the OFSM controller have shown that the OFSM controller achieve the control objectives more properly and efficiently.

2. Coupled dynamic equations

In this section, the coupled dynamic equations of a spacecraft with a partially-filled spherical fuel tank are described. Following [26], the spacecraft is modeled as a rigid body with mass m and moments of inertia (j_x, j_y, j_z) with respect to the spacecraft-fixed xyz -frame whose origin is at the tank center as shown in Fig. 1. Here F denotes the main thruster force, (δ_1, δ_2) are the gimbal deflection angles about (y, z) axes, and (M_x, M_y, M_z) denote the control moments about the spacecraft center of mass. The spacecraft is assumed to be moving in three-dimensional space with translational velocity (v_x, v_y, v_z) and angular velocity $(\Omega_1, \Omega_2, \Omega_3)$. The attitude of the spacecraft is given by the attitude angles $(\vartheta_1, \vartheta_2, \vartheta_3)$. The still liquid fuel is modeled as a rigid spherical mass m_0 with moment of inertia I_0 located at distance h_0 from the tank center. The

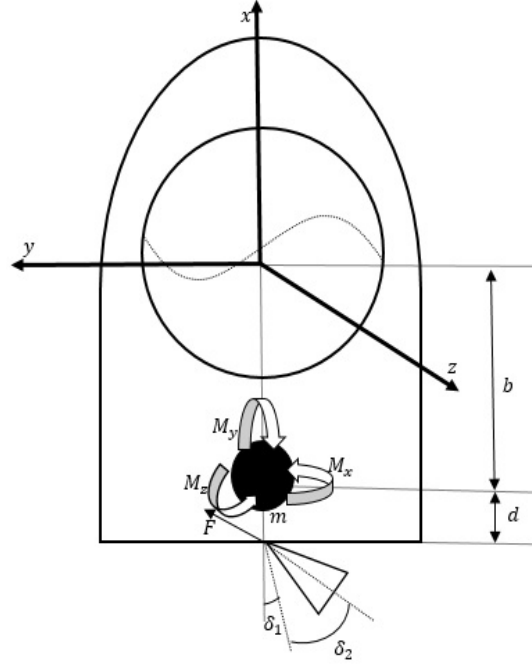


Figure 1: Model of the SC with a partially filled spherical fuel tank [26]

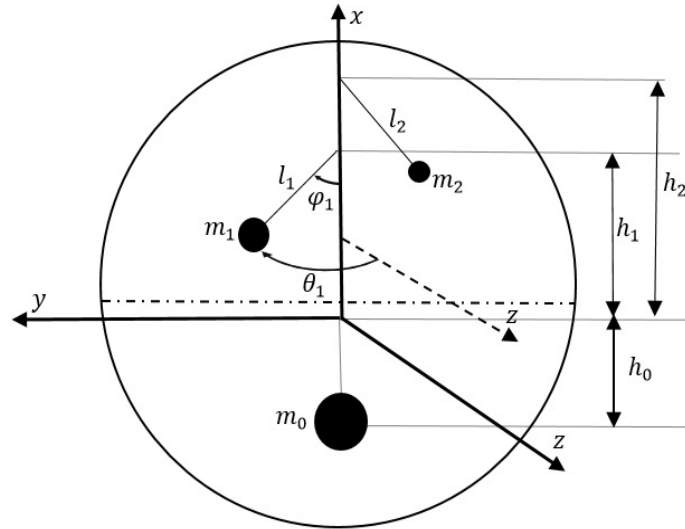


Figure 2: Model of sloshing and still liquid fuel in a spherical tank [26]

sloshing fuel is described by two spherical pendulums with masses (m_1, m_2) characterizing the first two most prominent sloshing modes. The pendulums of lengths (l_1, l_2) pivoted at (h_1, h_2) along x -axis have spherical coordinates (φ_i, θ_i) , $i = 1, 2$. After an input transformation from the original inputs $(M_x, M_y, M_z, \delta_1, \delta_2)$ to new control inputs $(u_1, u_2, u_3, u_4, u_5)$, the reduced equations of motion can be written as

$$a_x = \frac{F}{m + m_0 + \sum_{i=1}^2 m_i} \quad (1)$$

$$\dot{v}_y = u_1 - v_x \Omega_3 + v_z \Omega_1 \quad (2)$$

$$\dot{v}_z = u_2 - v_y \Omega_1 + v_x \Omega_1 \quad (3)$$

$$\dot{\Omega}_1 = u_3 \quad (4)$$

$$\dot{\Omega}_2 = u_4 \quad (5)$$

$$\dot{\Omega}_3 = u_5 \quad (6)$$

$$\begin{aligned} \ddot{\varphi}_i = & -\frac{1}{I_i + m_i l_i^2 \cos^2 \varphi_i} (I_i (\cos \theta_i (\dot{\Omega}_2) - \sin \theta_i (\dot{\Omega}_3 - \dot{\varphi}_i \sin \theta_i) - \dot{\theta}_i (\Omega_2 \sin \theta_i + \Omega_3 \\ & \cos \theta_i)) + m_i l_i (a_x \sin \varphi_i + \cos \varphi_i \sin \theta_i (a_y + \dot{\Omega}_3 h_i) + \cos \varphi_i \cos \theta_i (a_z \\ & - \dot{\Omega}_2 h_i) - \Omega_3 \sin \varphi_i (\Omega_3 h_i - \Omega_3 l_i \cos \varphi_i) + \Omega_2 \sin \varphi_i (-\Omega_2 h_i + \Omega_2 l_i \\ & \cos \varphi_i) + \Omega_1 \Omega_3 h_i \cos \varphi_i \cos \theta_i + \Omega_1 \Omega_2 h_i \cos \varphi_i \sin \theta_i) + \epsilon_i \dot{\varphi}_i), \quad i = 1, 2 \end{aligned} \quad (7)$$

$$\begin{aligned} \ddot{\theta}_i = & -\frac{1}{I_i} (I_i (-\dot{\Omega}_1 + \dot{\varphi}_i \Omega_2 \sin \theta_i + \dot{\varphi}_i \Omega_3 \cos \theta_i) + m_i l_i (\sin \varphi_i \cos \theta_i (a_y \\ & + \dot{\Omega}_3 h_i) - \sin \varphi_i \sin \theta_i (a_z - \dot{\Omega}_2 h_i) - \Omega_1 \Omega_3 h_i \sin \varphi_i \sin \theta_i + \Omega_1 \Omega_2 h_i \\ & \sin \varphi_i \cos \theta_i) + \epsilon_i \dot{\theta}_i), \quad i = 1, 2 \end{aligned} \quad (8)$$

where ϵ_i denotes the damping coefficient of the i th pendulum.

3. Design of optimum fuzzy sliding mode (OFSM) controller

Consider the reduced equations given by (1-8). In order to control the disturbed attitude of the spacecraft due to sloshing of its fuel, a fuzzy sliding mode (FSM) control approach using a PSO algorithm is introduced. This approach can stabilize all dynamic variables of the coupled spacecraft-fuel system. For designing a fuzzy sliding mode controller, first the following five sliding variables are defined.

$$S_1 = \left(\frac{d}{dt} + \mu_1\right)^{r_1} \int_0^t e_1(\tau) d\tau \quad (9)$$

$$S_2 = \left(\frac{d}{dt} + \mu_2\right)^{r_2} \int_0^t e_2(\tau) d\tau \quad (10)$$

$$S_3 = \left(\frac{d}{dt} + \mu_3\right)^{r_3} \int_0^t e_3(\tau) d\tau \quad (11)$$

$$S_4 = \left(\frac{d}{dt} + \mu_4\right)^{r_4} \int_0^t e_4(\tau) d\tau \quad (12)$$

$$S_5 = \left(\frac{d}{dt} + \mu_5\right)^{r_5} \int_0^t e_5(\tau) d\tau \quad (13)$$

where μ_{1-5} are the positive constants defining the bandwidth of the error dynamics and r_{1-5} are the relative degrees of attitude angles and velocity vectors of the spacecraft ($r_{1-2} = 1$, $r_{3-5} = 2$). Also, $e_1(\tau) = v_y - v_{yd}$, $e_2(\tau) = v_z - v_{zd}$, $e_3(\tau) = \vartheta_1 - \vartheta_{1d}$, $e_4(\tau) = \vartheta_2 - \vartheta_{2d}$, $e_5(\tau) = \vartheta_3 - \vartheta_{3d}$. In this paper, (v_{yd}, v_{zd}) and $(\vartheta_{1d}, \vartheta_{2d}, \vartheta_{3d})$ are set as zero. The time derivatives of the sliding variables $(\dot{S}_{1-5})$ can be expressed as

$$\dot{S}_1 = u_1 - v_x \Omega_3 + v_z \Omega_1 + \mu_1 v_y \quad (14)$$

$$\dot{S}_2 = u_2 - v_y \Omega_1 + v_x \Omega_2 + \mu_2 v_z \quad (15)$$

$$\dot{S}_3 = u_3 + 2\mu_3\vartheta_1 + \mu_3^2\Omega_1 \quad (16)$$

$$\dot{S}_4 = u_4 + 2\mu_4\vartheta_2 + \mu_4^2\Omega_2 \quad (17)$$

$$\dot{S}_5 = u_5 + 2\mu_5\vartheta_3 + \mu_5^2\Omega_3 \quad (18)$$

The inputs u_{1-5} are designed such that the following sliding conditions are satisfied:

$$\frac{1}{2} \frac{dS_{1-2}}{dt} \leq -K_{S_{1-2}} |S_{1-2}| \quad (19)$$

$$\frac{1}{2} \frac{dS_{3-5}^2}{dt} \leq -K_{S_{3-5}} |S_{3-5}| \quad (20)$$

where $K_{S_{1-5}}$ are arbitrary positive constants. Consequently, the control inputs (u_{1-5}) are obtained as

$$u_1 = v_x\Omega_3 - v_z\Omega_1 - \mu_1v_y - K_{S_1}sgn(s_1) \quad (21)$$

$$u_2 = v_y\Omega_1 - v_x\Omega_2 - \mu_2v_z - K_{S_2}sgn(s_2) \quad (22)$$

$$u_3 = -2\mu_3\Omega_1 - \mu_3^2\vartheta_1 - K_{S_3}sgn(s_3) \quad (23)$$

$$u_4 = -2\mu_4\Omega_2 - \mu_4^2\vartheta_2 - K_{S_4}sgn(s_4) \quad (24)$$

$$u_5 = -2\mu_5\Omega_3 - \mu_5^2\vartheta_3 - K_{S_5}sgn(s_5) \quad (25)$$

where sgn stands for the signum function. In order to mitigate the undesirable effects of chattering the following continuous saturation function ($sat(S_{1-5})$) is

employed instead of the signum function:

$$sat(S_{1-5}) = \begin{cases} -1 : & \frac{S_{1-5}}{\Gamma_{1-5}} \leq -1 \\ \frac{S_{1-5}}{\Gamma_{1-5}} : & -1 < \frac{S_{1-5}}{\Gamma_{1-5}} \leq 1 \\ 1 : & \frac{S_{1-5}}{\Gamma_{1-5}} > 1 \end{cases} \quad (26)$$

where Γ_{1-5} are the thin boundary layer widths. Now, a fuzzy gain scheduling is proposed to exploit fuzzy Mamdani rules and generate sliding mode controller parameters. For this purpose, it is assumed that $K_{S_{1-5}}$ are bounded in $K_{S_{1-5}min}$ and $K_{S_{1-5}max}$ ($[K_{S_{1-5}min} \quad K_{S_{1-5}max}]$); the boundaries are derived experimentally as

$$\begin{aligned} K_{S_{1-5}min} &= 0.5K_{S_{1-5}}, \\ K_{S_{1-5}max} &= 2K_{S_{1-5}} \end{aligned} \quad (27)$$

The sliding mode controller parameters $K_{S_{1-5}}$ are determined as

$$K_{S_{1-5}} = K_{S_{1-5}min} + (K_{S_{1-5}max} - K_{S_{1-5}min})K'_{S_{1-5}} \quad (28)$$

where $K'_{S_{1-5}}$ are outputs of fuzzy Mamdani logic. As it is indicated in Fig. 3, each output has two Gaussian membership functions (Big and Small) in the range of $[0 \quad 1]$. Also, as depicted in Fig. 4, each of dynamic variables of this fuzzy logic has five triangle (NM, NS, ZO, PS and PM) and two trapezoidal (NB and PB) membership functions in the allowed range of dynamic variables. For determining the fuzzy rules, the sign of saturation function of sliding surfaces are considered as other fuzzy inputs (α). Each one of these inputs has two trapezoidal membership functions (see Fig. 5).

In order to define rules for determining the outputs $K'_{S_{1-5}}$, a novel gain scheduling is introduced. This scheduling is given in Table 1.

3.1. PSO algorithm

Now for optimizing the lower and upper bounds of control gains used in the fuzzy logic, in order to minimize the control inputs, a PSO algorithm is em-

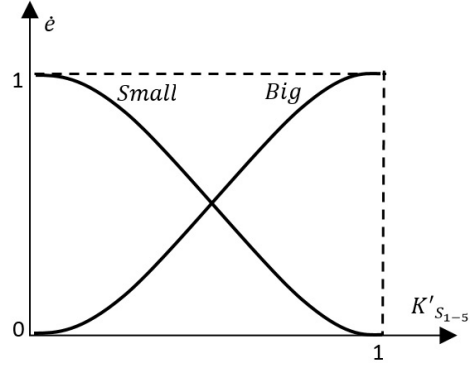


Figure 3: Gaussian membership functions of Mamdani logic outputs

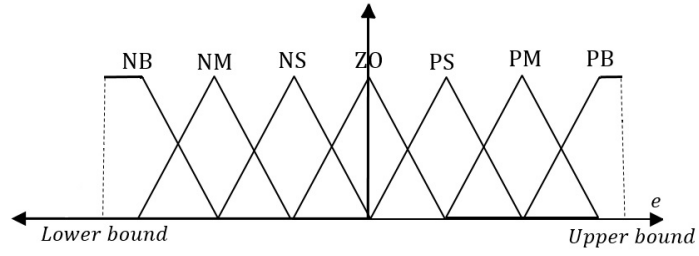


Figure 4: Membership functions of input dynamic variables

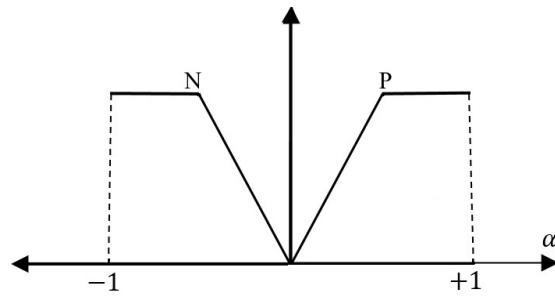


Figure 5: Membership functions of saturation function of sliding surfaces

Table 1: Rules for defining K'_{S1-5}

α		P	N
e	NB	Small	Big
	NM	Small	Big
	NS	Big	Small
	ZO	Big	Small
	PS	Big	Small
	PM	Small	Big
	PB	Small	Big

ployed. So, function that has to be minimized (MF) is formulated in following equation.

$$MF = \int_{t=0}^T (M_x + M_y + M_z + \delta_1 + \delta_2) dt \quad (29)$$

PSO is a computational technique which starts with random selections known as particles. Introduced particles ($\mathbf{P}_j(\mathbf{it})$) shift toward the best solution with random velocity ($\mathbf{V}_j(\mathbf{it})$) iteratively (j stands for number of particles). The next position of the particles depends on their local best and global best positions in search space. Also, these positions will be updated in each iteration as better positions are calculated for the other particles.

$$\mathbf{P}_j(\mathbf{it}) = [p_{j_1}(it), \dots, p_{j_m}(it)] \quad (30)$$

$$\mathbf{V}_j(\mathbf{it}) = [v_{j_1}(it), \dots, v_{j_m}(it)] \quad (31)$$

where m denotes the number of parameters at it th iteration. At each iteration there are n positions for each particle, therefore there are n velocity vectors for each particle.

$$\mathbf{PoP}(\mathbf{it}) = [P_1(it) \quad \dots \quad P_n(it)]^T \quad (32)$$

$$\mathbf{V}(\mathbf{it}) = [V_1(it) \quad \cdots \quad V_n(it)]^T \quad (33)$$

The determined local best in the PSO algorithm is the best position, which is calculated by movement of the particles through the search space. Therefore, if $P_j(it) < P_j(\tau)$ ($\tau = 0, 1, \dots, it$), then $P_j^{lbest} = P_j(it)$, and the vector of local best positions will be defined as follows.

$$\mathbf{P}_j^{lbest}(\mathbf{it}) = [p_{j_1}^{lbest}(it) \quad \cdots \quad p_{j_m}^{lbest}(it)]^T \quad (34)$$

Also, if $P_j(it) < P_j^{lbest}(it)$ ($j = 1, 2, \dots, Pop$), then the global best position will be defined as $P^{gbest} = P_j^{lbest}$ and consequently, global best velocity vector will be expressed as

$$\mathbf{P}^{gbest}(\mathbf{it}) = [p_1^{gbest}(it) \quad \cdots \quad p_m^{gbest}(it)]^T \quad (35)$$

In the PSO algorithm, a parameter known as inertia weight ($W_I(it)$) is defined to control the impact of previous velocities in the current velocity, and will be updated at each iteration by the following equation.

$$W_I(it) = \gamma W_I(it - 1) \quad (36)$$

where γ denotes damping ratio of inertia weight. So, j th particle's velocity will be updated as follows.

$$\begin{aligned} v_{j_k}(it) = & W(it)v_{j_k}(it - 1) + c_1g_1(p_{j_k}^{lbest}(it - 1) - p_{j_k}(it - 1)) \\ & + c_2g_2(p_k^{gbest} - p_{j_k}(it - 1)) \end{aligned} \quad (37)$$

Here g_1 and g_2 stand for personal and global learning coefficients, respectively. Furthermore, c_1 and c_2 are uniformly disturbed random numbers in range of zero and one. As a result, the position of each particle will be updated as below.

$$p_{j_k}(it) = v_{j_k}(it) + p_{j_k}(it - 1) \quad (38)$$

The flowchart of this algorithm is illustrated in Fig. 6. In this figure, $max(it)$ represents the maximum number of iterations of the PSO algorithm.

4. Design of optimum SDRE controller

In this section, an SDRE-based controller is designed [5, 27], the performance of which is compared to that of the PSO-based controller in the subsequent section. The system (2)-(6) can be rewritten in the following quasi-linear form:

$$\dot{x} = A(x)x + Bu \quad (39)$$

where $x = [v_y \ v_z \ \vartheta_1 \ \Omega_1 \ \vartheta_2 \ \Omega_2 \ \vartheta_3 \ \Omega_3]^T$ is the state vector, and the matrices $A(x)$ and B are defined as

$$A(x) = \begin{bmatrix} 0 & \Omega_1 & 0 & 0 & 0 & 0 & 0 & -v_x \\ -\Omega_1 & 0 & 0 & 0 & 0 & v_x & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (40)$$

The cost index J that is to be minimized is given by

$$J = \frac{1}{2} \int_0^\infty [x^T Q(x)x + u^T R(x)u] dt \quad (41)$$

where $Q(x)$ and $R(x)$ are positive definite matrices. The optimal feedback control is expressed as

$$u = -R^{-1}(x)B^T P(x)x \quad (42)$$

where $P(x)$ is the positive definite solution of the state-dependent Riccati equation

$$A^T(x)P(x) + P(x)A(x) - P(x)BR^{-1}(x)B^T P(x) + Q(x) = 0 \quad (43)$$

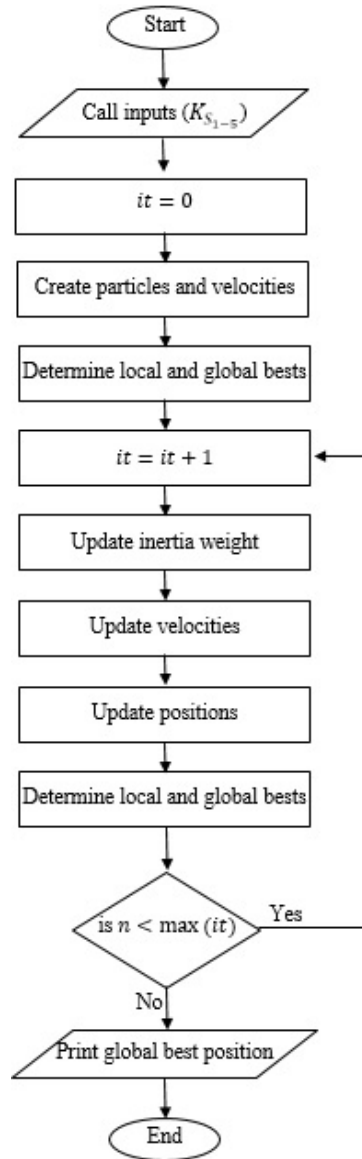


Figure 6: Flowchart of PSO algorithm

5. Numerical results

In order to illustrate the performance of the PSO-based OFSM controller designed in Section 3, a computer simulation is conducted. The physical parameters of the coupled spacecraft-fuel system introduced in Section 2 are given in Table 2. The initial conditions are presented in Table 3.

Table 2: Physical parameters of the system

Parameter	Value	Unit	Parameter	Value	Unit
m	975	kg	l_1	0.2	m
j_x	600	$kg.m^2$	l_2	0.1	m
j_y	800	$kg.m^2$	h_0	0.05	m
j_z	400	$kg.m^2$	h_1	0.6	m
m_0	480	kg	h_2	0.9	m
I_0	75	$kg.m^2$	b	-0.6	m
m_1	50	kg	d	1.2	m
m_2	5	kg	F	2450	N
I_1	10	$kg.m^2$	ϵ_1	3.7	$kg.m^2/s$
I_2	1	$kg.m^2$	ϵ_2	0.5	$kg.m^2/s$

Table 3: Initial conditions

Variable	Value	Unit	Variable	Value	Unit
v_{x_0}	3000	m/s	φ_{1_0}	30	degree
v_{y_0}	75	m/s	φ_{2_0}	40	degree
v_{z_0}	25	m/s	θ_{1_0}	20	degree
ϑ_{1_0}	5	<i>Degree</i>	θ_{2_0}	10	degree
ϑ_{2_0}	2	<i>Degree</i>	$\dot{\vartheta}_{1_0} = \dot{\vartheta}_{2_0} = \dot{\vartheta}_{3_0}$	0	rad/s
ϑ_{3_0}	-5	<i>Degree</i>	$\dot{\varphi}_{1_0} = \dot{\varphi}_{2_0} = \dot{\theta}_{1_0} = \dot{\theta}_{2_0}$	0	rad/s

The control parameters of the fuzzy sliding mode controller and the PSO

parameters are given in Table 4 and Table 5, respectively. By applying the PSO algorithm for minimizing the control inputs (M_x , M_y , M_z , δ_1 and δ_2), the optimal gains of the fuzzy sliding mode controller are obtained. As illustrated in Figs. 7-11, the PSO algorithm can find the optimal gains efficiently.

Table 4: Parameters of sliding mode controller

Control parameter	Value	Cotrol parameter	Value
$\mu_1 = \mu_2$	2×10^{-2}	$\mu_3 = \mu_4 = \mu_5$	2×10^{-2}
$\Gamma_1 = \Gamma_2$	2	$\Gamma_3 = \Gamma_4 = \Gamma_5$	0.2

Table 5: PSO parameters

Parameter	Value	Parameter	Value
n	12	γ	0.99
$max(it)$	120	g_1	2
W_I	1	g_2	2

The gains of the fuzzy sliding mode controller for applying in PSO algorithm are bounded between $[1 \times 10^{-4} \quad 1 \times 10^{-7}]$ for $K_{S_{1,2}}$ and $[5 \times 10^{-5} \quad 5 \times 10^{-8}]$ for $K_{S_{3-5}}$. The gains converge to their optimal values after 100 iterations. The values for $K_{S_{1-5}}$ are 1×10^{-4} , 1×10^{-7} , 5×10^{-8} , 5×10^{-5} and 5×10^{-5} , respectively. In addition, Fig. 12 shows the best cost versus Number of Function Evaluation (NFE).

The optimal gains determined by the PSO algorithm are used as $K_{S_{1-5}}$ (introducing the lower and upper bounds in the fuzzy logic). The time responses of the dynamic variables of the system are shown in Figs. 13-20. As Fig. 13 shows, the transverse velocities of the spacecraft (v_y , v_z) converge to zero slightly after 200 seconds.

Also, the attitude angles of the spacecraft (ϑ_1 , ϑ_2 , ϑ_3) asymptotically tend to zero after 300, 250 and 250 seconds, respectively. Moreover, as mentioned before,

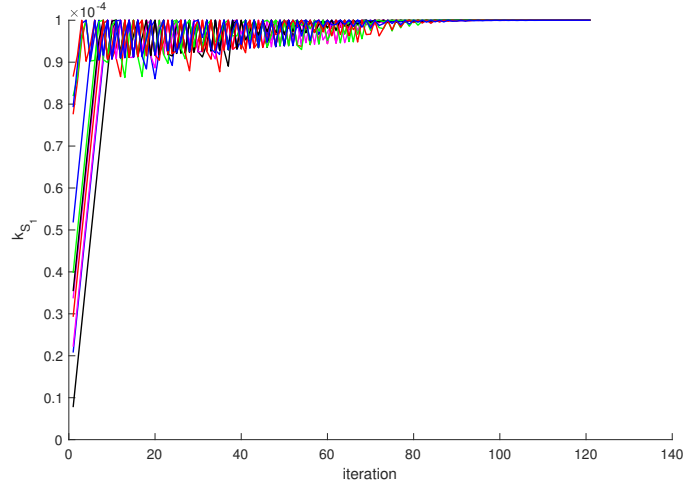


Figure 7: Convergence of K_{S_1}

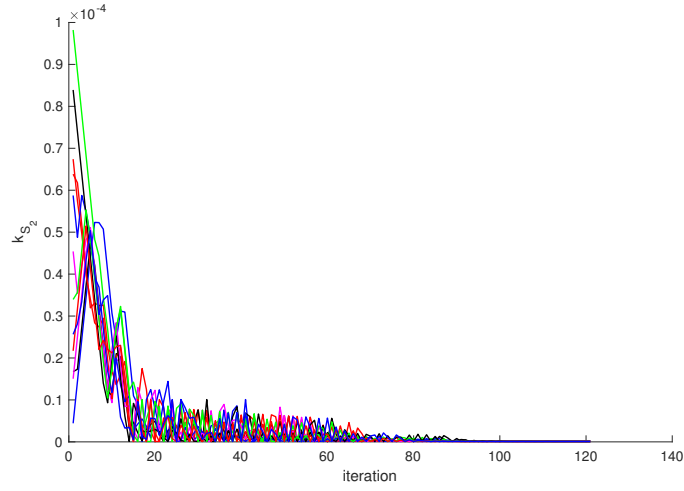


Figure 8: Convergence of K_{S_2}

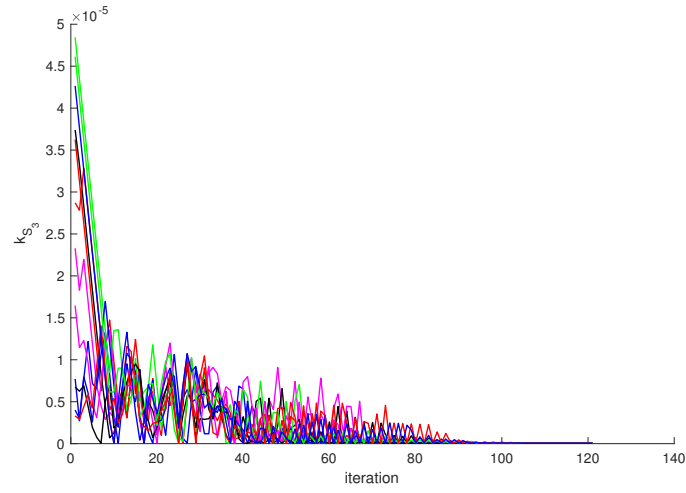


Figure 9: Convergence of K_{S_3}

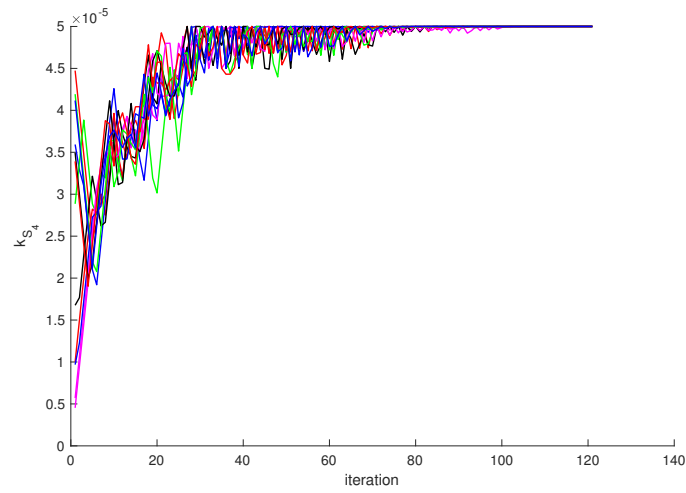


Figure 10: Convergence of K_{S_4}

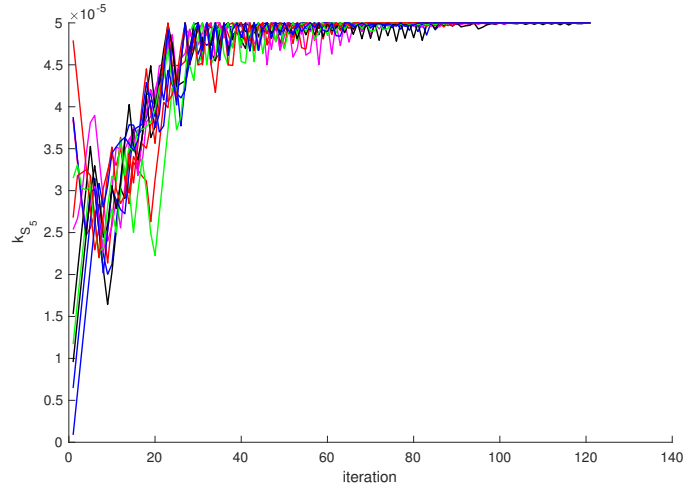


Figure 11: Convergence of K_{S_5}

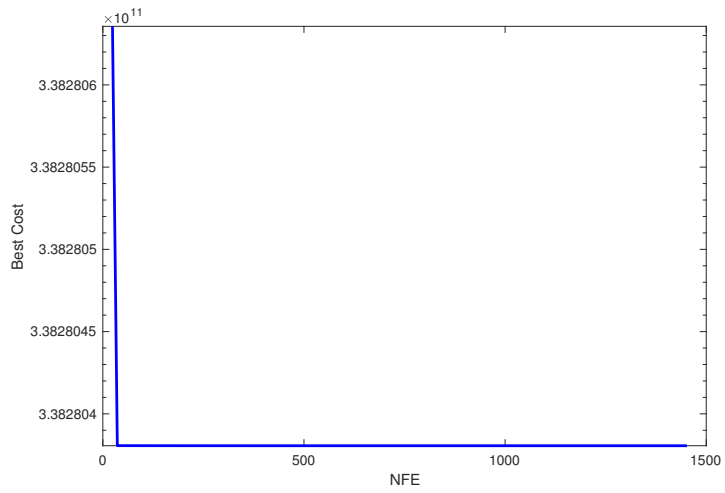


Figure 12: Best cost versus NFE

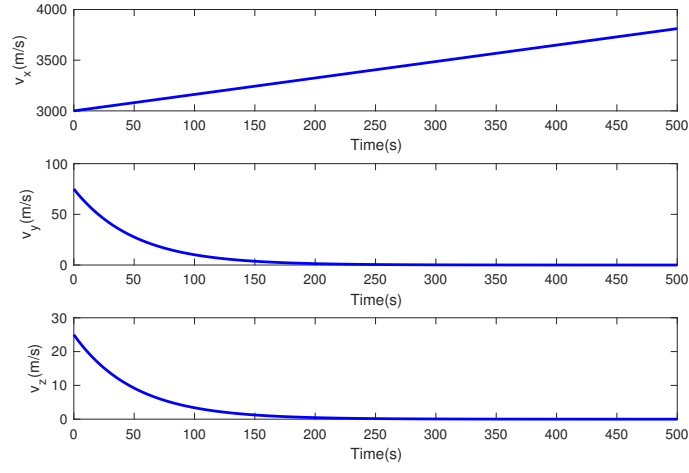


Figure 13: Time responses of spacecraft velocities

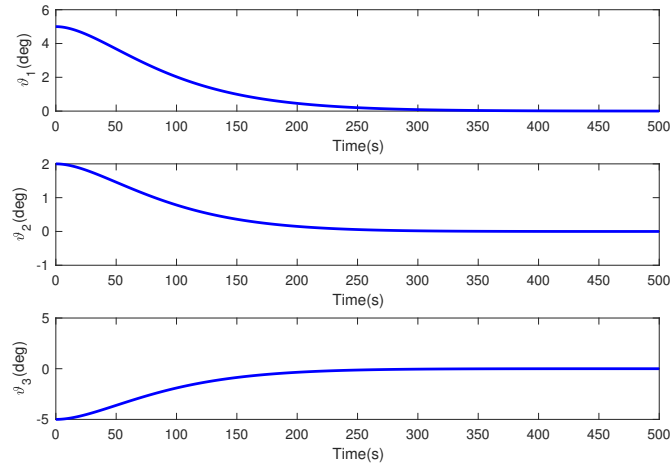


Figure 14: Time responses of spacecraft attitude angles

sloshing dynamics are coupled with the spacecraft dynamics, so the designed controller can stabilize both of them simultaneously. The time responses of the spherical angles of the pendulums are illustrated in Figs. 15-16. These figures show that the pendulum angles about x axis stabilize to zero and the angles about y axis tend to their stable positions after 400 and 50 seconds, respectively.

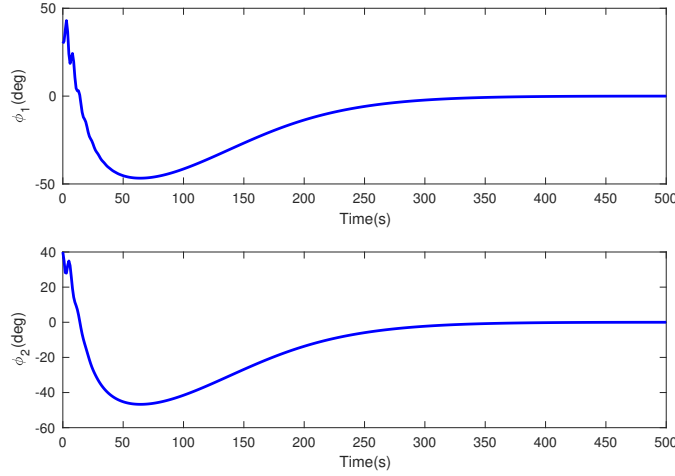


Figure 15: Time responses of φ_i

The time behaviors of the gimbal angles are shown in Fig. 17. Fig. 18 indicates the time behaviors of torques about center of the spacecraft mass. The variations of the fuzzy outputs are illustrated in Figs. 19-20. As simulation results show, the novel OFSM controller with the optimized gains created by the PSO algorithm achieves the control goals efficiently.

Now a comparison between the designed controllers in Section 3 and Section 4 is included in Figs. 21-24 to illustrate the effectiveness and advantages of the OFSM controller. As seen in Figs. 21-22, the time responses of the spacecraft dynamic variables (except ϑ_1) tend to zero after the same time elapse, whereas overshoots caused by the SDRE controller are drastic and undesirable. Also Figs. 23-24 show other advantages; variations of gimbal deflection angles using the OFSM controller are more smooth than the SDRE controller. Moreover

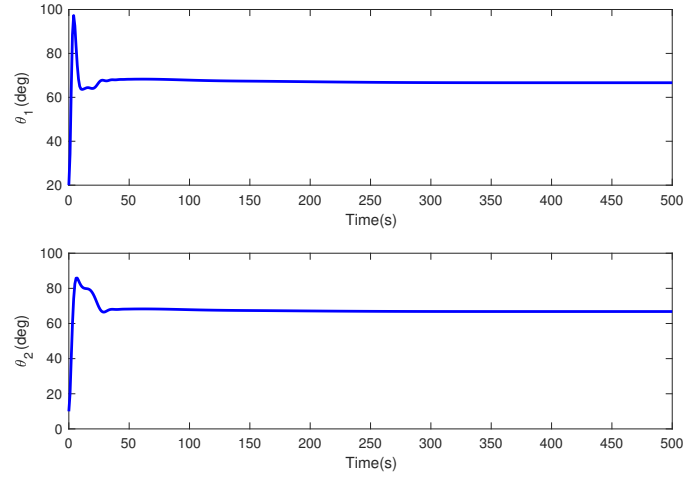


Figure 16: Time responses of θ_i

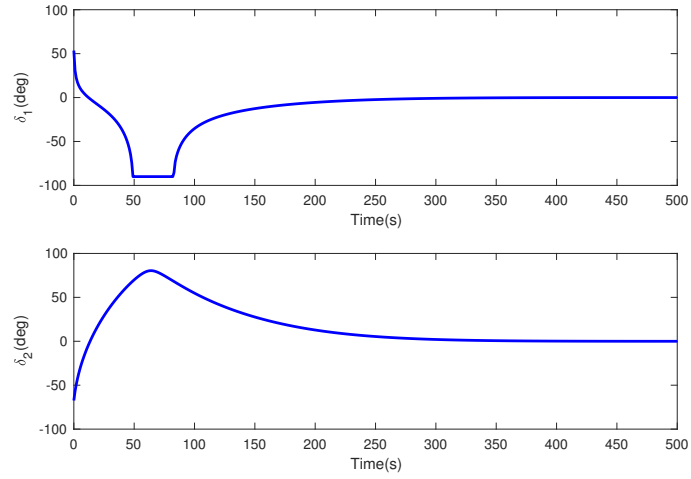


Figure 17: Gimbal deflection angles δ_1 and δ_2

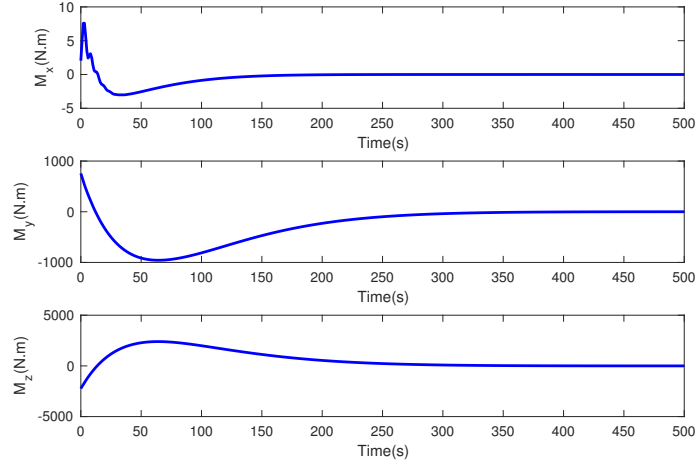


Figure 18: Control moments M_x , M_y and M_z

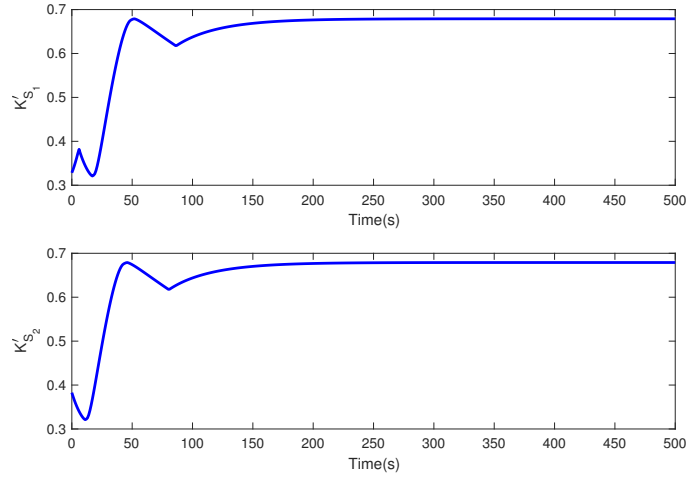


Figure 19: Time history of $K'_{S1,2}$

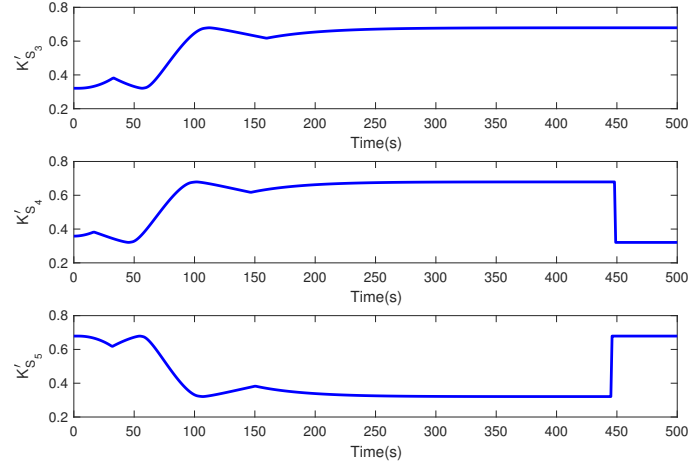


Figure 20: Time history of $K'_{S_{3-5}}$

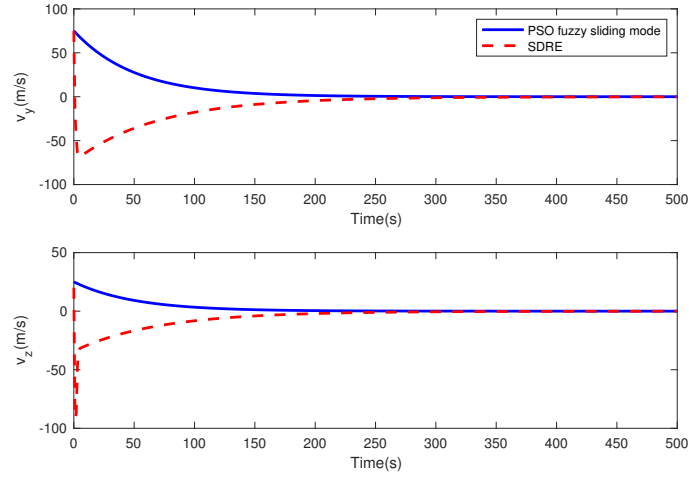


Figure 21: Comparison of spacecraft velocities

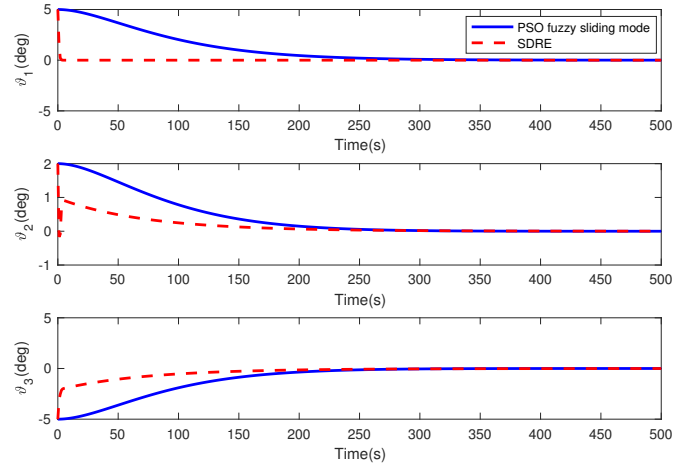


Figure 22: Comparison of spacecraft attitude angles

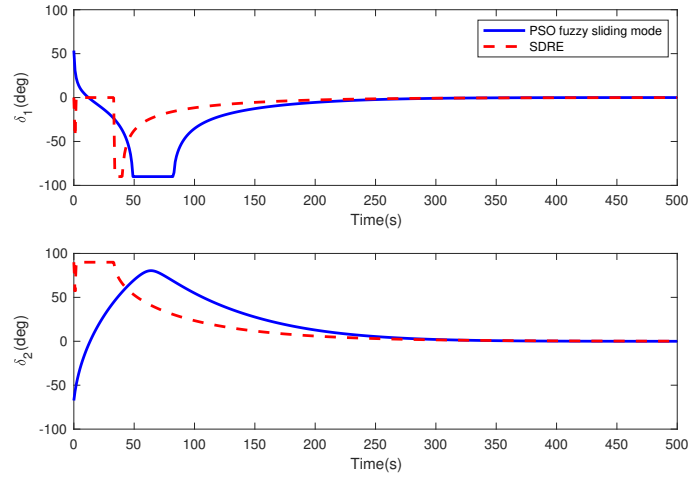


Figure 23: Comparison of gimbal deflection angles

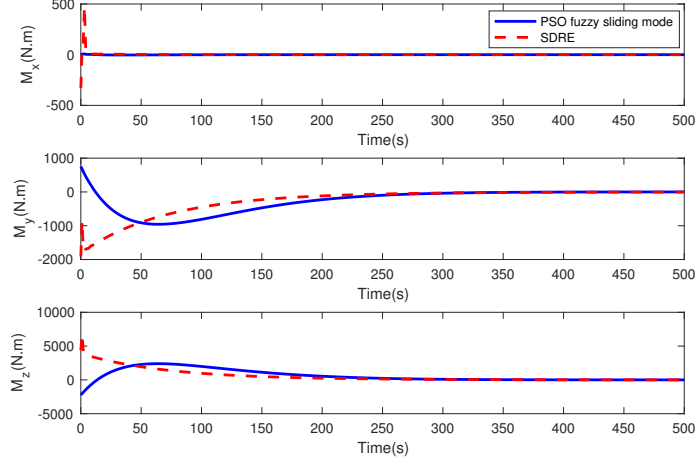


Figure 24: Comparison of control moments

the SDRE controller requires higher control moments than those of the OFSM controller.

6. Conclusion

An optimal fuzzy sliding mode controller is presented for the control of coupled spacecraft-fuel system. The proposed fuzzy controller is defined by a novel gain scheduling scheme, where the fuzzy gains are generated on-line based on the dynamic system variables and the sign of saturation function of sliding surfaces. A PSO algorithm is employed to optimize the fuzzy gains in order to achieve optimal controller performance with minimum control inputs. Computer simulations are included to illustrate the effectiveness of the controller. The performance of the PSO-based controller is compared to that of the SDRE-based controller.

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